

Supplementary Material for Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

Anonymous Submission

Scope of the Supplement

This anonymous supplementary material contains reproducibility and audit details that are intentionally kept out of the main technical-content pages. These tables support the evidence-condition controller evaluation but are not used as additional main effectiveness evidence. In particular, the workflow pilot is a workflow-shape/interface validation only, and the scalar singleton/Chao proxy is a negative control because it does not encode source-route evidence conditions.

Reproducibility Notes

Script entry points. All paths are relative to the repository root. The supplement and paper figures are assembled from `2/analysis/research_object_geometry/real_agent_pilot/` by running the Python entry point `credibility_supplement/run_credibility_supplement.py`. The frozen task logs used by this assembler are produced by `scripts/run_blind_policy_task.py`, `scripts/run_blind_code_task.py`, `external_validation_requests/run_external_requests_validation.py`, `controller_validation_v1/run_controller_validation_v1.py`, `controller_validation_v1/run_controller_validation_v2.py`, and `external_validation_v2/run_external_validation_v2.py`. The controlled stress tests are in `2/analysis/research_object_geometry/`: `controlled_geometry_simulation_v2.py` and `controlled_exposure_localization_v3.py`.

Seeds and thresholds. Generated-task method validation, external repository repair validation, and external boundary validation use 200 fixed seeds per challenger or controller policy before oracle scoring. Safe-state validation uses 6000 order/budget perturbations. The controller-validation thresholds are $\tau_s = 0.75$ for source-route support and $\tau_g = 0.70$ for exposure Gini. The 0.90 bounded-oracle recall threshold is used only for post-hoc false-certification evaluation and is not visible to the controller.

Data and log format. Primary run logs are JSONL action/evidence ledgers. Each event records a task identifier, source family, source-route stratum, route, action or scan record, discovered item identifier when present, new-item flag, and cost fields used for post-stop repair or confirmation-audit accounting. Aggregated CSVs in `credibility_supplement/results/` contain condition summaries, controller decisions, repair confidence intervals, seeded safe-state summaries, threshold/budget sweeps, oracle summaries, route-pattern examples, and leakage controls.

Oracle CSV and JSONL fields. Oracle item files use the fields `task_id`, `item_id`, `oracle_label`, `oracle_bucket`, `source_path`, `source_family`, `source_route_stratum`, `reportable`, and, for external repository JSONL files, a representative line. Source-route count CSVs use `source_family`, `oracle_bucket`, and `oracle_items`. Oracle labels and oracle totals are loaded only after trajectories, repair orderings, and challenger choices are fixed.

Figure generation. The same supplement assembler writes `main_results_overview.png`, `controller_decision_matrix.png`, and `repair_sensitivity_summary.png` to `2/paper/figures/`. Run the assembler above, then compile from `2/paper/` with `latexmk -pdf -interaction=nonstopmode -outdir=build` and either `evidence_condition_geometry_aaai_v2.tex` or `supplementary_material.tex`.

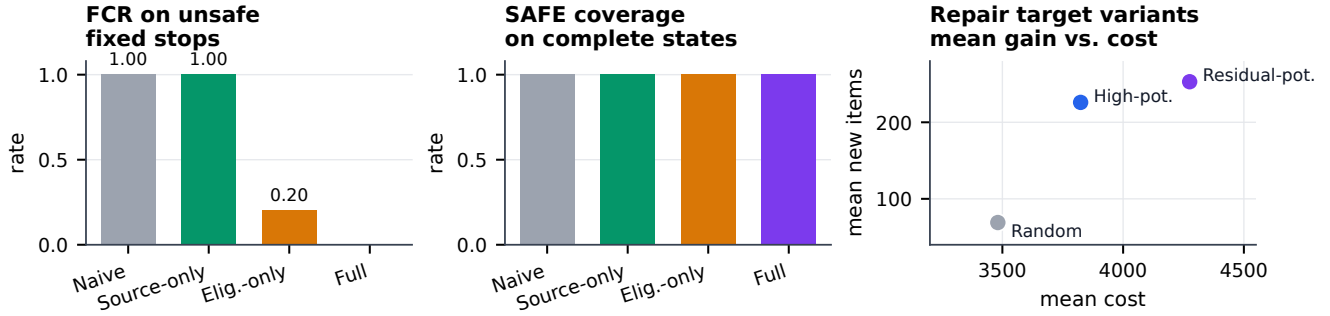


Figure 1: Controller variant comparison. Naive stop acceptance and source-only control accept unsafe observed stop states. Source-route eligibility reduces false SAFE decisions but remains insufficient on the broad-but-productive `urllib3` boundary. Repair-enabled variants preserve safe-state coverage; residual-potential is the method instance used by the full controller.

Table 1: Negative control: scalar singleton/Chao proxy on homogeneous runs. The proxy observes item-count sparsity but does not encode source-route condition, so it is not a competitive controller baseline for certificate mismatch.

Task	Observed	Oracle total	Recall	f_1	f_2	f_1/obs	Chao1	Scalar stop	False if stop
<code>policy_docset_v1</code>	17	24	0.708	17	0	1.000	153.0	No	No
<code>code_repo_v1</code>	6	20	0.300	6	0	1.000	21.0	No	No
<code>requests</code>	31	298	0.104	31	0	1.000	496.0	No	No
<code>urllib3</code>	135	699	0.193	135	0	1.000	9180.0	No	No

Controller Variant Details

Figure 1 compares decision policies on the same fixed bounded-audit states used in the main paper. This is a policy ablation, not a new task suite: oracle labels are used only after decisions are fixed to compute false-SAFE rate, safe-state coverage, decision rates, and repair gain/cost.

Computing environment. The checked environment is Windows 11 with Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, and Matplotlib 3.10.6. The code uses only local frozen snapshots for the external repository audits; no network access is required to reproduce the reported tables and figures from the included logs and snapshots.

AAAI Reproducibility Checklist

This paper includes computational experiments. The main text states the bounded scope, assumptions, metrics, thresholds, and controller rule. This supplement records script entry points, seeds, data/log formats, oracle fields, figure-generation commands, software environment, confidence intervals, and threshold/budget sweeps. Randomness is controlled by fixed seed sets before oracle scoring; oracle labels, oracle totals, post-hoc recall, and undiscovered true-item counts are used only for post-hoc scoring. The implementation and generated artifacts are organized as a code/data supplement for anonymous review and can be released on publication.

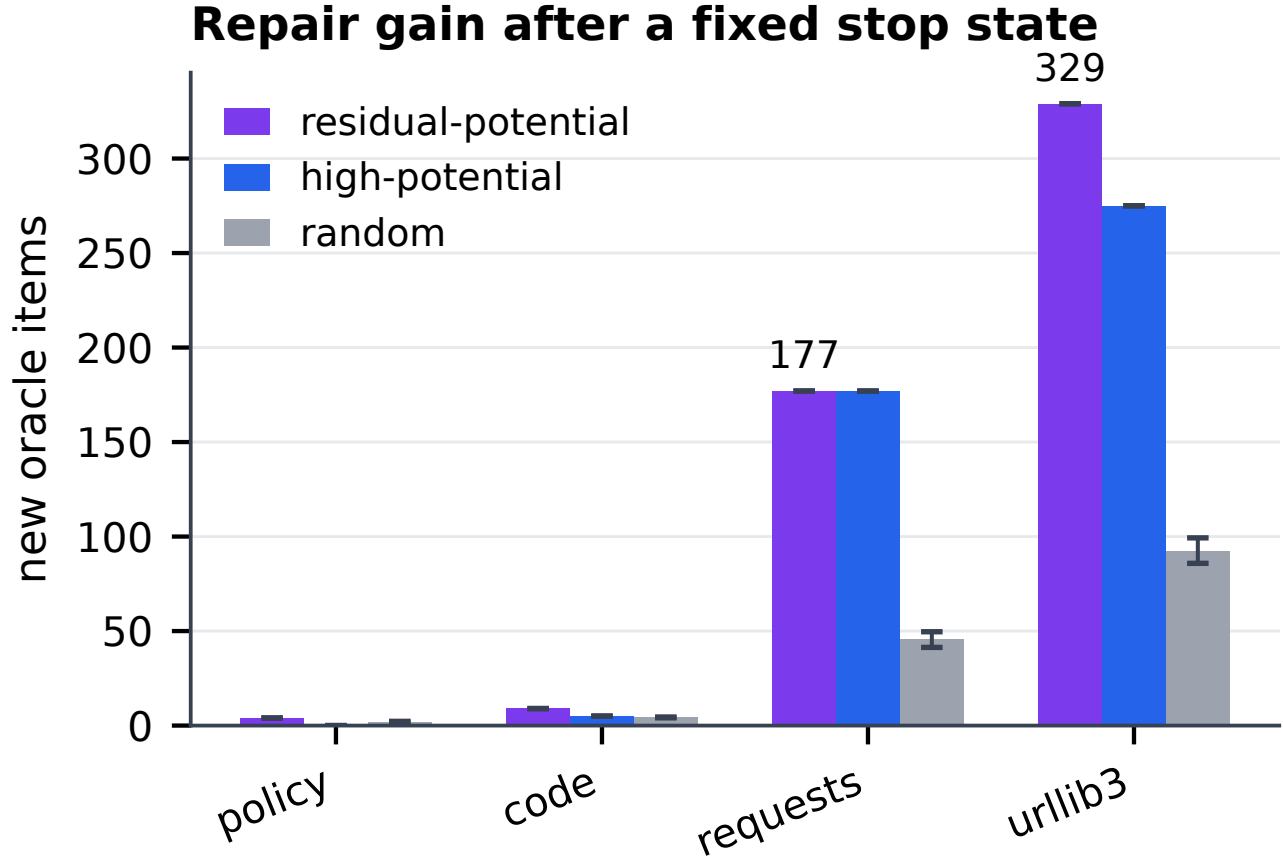


Figure 2: Task-level repair gain for residual-potential, high-potential, and random probes with 95% confidence intervals. This detail complements the main paper’s controller-variant comparison and is not a separate task expansion.

Table 2: Sensitivity sweep numeric summary. Ranges aggregate threshold settings or repair-budget settings within each external task.

Task	Sweep	SAFE rate	CONTINUE rate	ABSTAIN rate	Max FCR	Mean repair cost
requests	threshold	0.000-0.000	0.995-1.000	0.000-0.005	0.000	2854.6
urllib3	threshold	0.000-0.000	0.990-1.000	0.000-0.010	0.000	4831.8
requests	budget	0.000-0.000	0.635-1.000	0.000-0.365	0.000	2966.4
urllib3	budget	0.000-0.000	0.640-1.000	0.000-0.360	0.000	4036.5

Table 3: External oracle construction summary. Pattern-defined external oracles are line-level source-route evidence occurrences from frozen local snapshots.

Repo	Version	Items	Routes	Nonempty strata	Granularity
requests	2.32.5	298	4	18	line-level source-route
urllib3	2.5.0	699	5	24	line-level source-route

Table 4: Small logged workflow pilot for workflow-shape/interface validation only. The pilot fixes the prompt and records independent contexts, action/evidence events, localized stop proposals, and the controller-facing decision record; it is not used as main effectiveness evidence.

Task	Model	Agents	Independent contexts	Action events	Evidence events	Stop proposals	Controller decision
T_doc_dynamic_workflow_smoke	logged LLM-agent workflow	3	3	39	35	3	not scored; workflow-shape validation

Table 5: Seeded safe-state validation. These are oracle-safe broad-complete states with seeded order perturbations; the controller should remain SAFE rather than over-reacting to extra repair opportunity.

Task	State	Order/budget sweep	Runs	#SAFE	#CONTINUE	#ABSTAIN	FCR	Safe cov.	Repair gain	Mean cost	Cost range
requests	route_partitioned	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3041.2	157-7647
urllib3	extended_audit	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3678.2	275-9072

Table 6: Route-match consistency sanity check for external oracle positives. A small sample of line-level positives was checked for route consistency; ambiguous lines are retained as such rather than treated as full validation.

Repo	Route	Sample	Positive	Precision	Ambiguous	Notes
requests	compat_route	8	8	1.000	0	line-level oracle-positive sample
requests	exception_route	8	8	1.000	0	line-level oracle-positive sample
requests	timeout_route	8	8	1.000	0	line-level oracle-positive sample
requests	tls_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	cleanup_route	8	7	0.875	1	line-level oracle-positive sample
urllib3	exception_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	retry_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	timeout_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	tls_route	8	8	1.000	0	line-level oracle-positive sample

Table 7: Leakage control for controller and repair decisions. Runtime decisions use only visible evidence. Oracle labels, oracle totals, post-hoc recall, and undiscovered true item counts are used only after trajectories and challenger choices are fixed.

Component	Runtime allowed	Oracle forbidden
Exposure	visits, scans, actions	missing mass
Support/Gini	exposure counts	post-hoc recall
Potential	text, routes, matches	hidden true items
Decision	condition and repair	target distribution

Table 8: External oracle appendix detail. Each route uses a frozen regex pattern; items are deduplicated by source, route, and line. To avoid unreadable regex truncation, the appendix reports a short pattern category and one representative line-level item per route; full regex rules and items are in the generated CSVs.

Repo	Route	Items	Pattern category	Dedup	Representative item
requests	tls_route	102	TLS/certificate/verification terms	source+route+line	adapters:tls_route:103: # cert path
requests	timeout_route	31	timeout/connect/read-timeout terms	source+route+line	adapters:timeout_route:120: self, request, stream=False, timeout=None, verify=True, cert=None, proxies=None
requests	exception_route	121	exception/raise/error terms	source+route+line	adapters:exception_route:136: raise NotImplementedError
requests	compat_route	44	compatibility/deprecation terms	source+route+line	adapters:compat_route:32: from .compat import basestring, urlparse
urllib3	timeout_route	135	timeout/connect/read-timeout terms	source+route+line	connection:timeout_route:137: timeout: .TYPE.TIMEOUT = _DEFAULT.TIMEOUT,
urllib3	retry_route	168	retry/backoff/status terms	source+route+line	connectionpool:retry_route:1011: retries,
urllib3	tls_route	111	TLS/certificate/verification terms	source+route+line	connection:tls_route:1005: cert,
urllib3	exception_route	170	exception/raise/error terms	source+route+line	connection:exception_route:1015: except BaseException:
urllib3	cleanup_route	115	close/release/finally terms	source+route+line	connection:cleanup_route:1016: ssl_sock.close()